Hazard Analysis
Lake

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Table 1: Revision History

Date	**Developer(s)**	**Change**
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1 Introduction

This document is the hazard analysis for Lake. Lake splits a large language model (LLM) into blocks of layers and runs each block on a different device on a local network, so that the devices can run a model together that none of them could run alone. The leader node's coordinator assigns blocks of layers to the devices and accepts prompts from the user. Each follower node runs its block and passes the result on to the next device.

1.1 Definition of a Hazard

For this document, a hazard is a state or condition of Lake that, combined with conditions in its environment that the system does not control, can lead to one of the losses listed in Section 2.3. Examples of such conditions are a device losing power, a congested network, or an untrusted device on the same network. For example, a device being assigned more layers than its memory can hold is a hazard, and the crashes that follow are the loss. A failure mode is a specific way a component can fail, and it can create a hazard.

Lake has almost no potential for physical harm, so its hazards are mainly about privacy, security, the correctness and availability of responses, and the effect of the system on the user's own devices.

2 Scope and Purpose of Hazard Analysis

2.1 Purpose

The purpose of this analysis is to find the ways Lake can cause a loss while the project is still in the requirements phase, when a hazard can be removed by changing a requirement instead of the code. Each hazard found in the FMEA leads to a safety or security requirement, which is added to the SRS and then verified through the V&V Plan.

2.2 Scope

In scope:

- The software components of Lake: the coordinator, the model partitioner, the inference engine on each device, the communication between devices, and the user interface.
- The interaction of these components with the user's devices, the local network that connects them, and the model files that are loaded.

Out of scope:

- The content the model generates. Harmful, biased or false text produced by a correctly running model is a property of the model, which the user chooses and Lake treats as an input. Incorrect output caused by Lake itself is in scope.
- Defects in the operating system, drivers or GPU hardware. These are outside Lake's control, so this analysis does not try to prevent them. It does cover their effects, such as a device crashing partway through a request, because Lake must respond to them.
- Attackers outside the local network, and physical access to a device. Lake is not exposed to the internet, and a person with physical access to a device already controls everything on it.

2.3 Losses

The hazards in this document can lead to the losses below.

ID	Loss	Description
L1	Loss of privacy or security	Prompts, responses or data passed between devices are seen by a person or device that is not meant to see them, or an untrusted device gives an attacker access to data or code on the user's devices.
L2	Loss of output integrity	The user receives a wrong or incoherent response because of Lake, for example from data corrupted between devices or a wrong assignment of layers, without knowing that it is wrong.
L3	Loss of availability	A request fails or never completes, or a session is lost partway through generating a response.
L4	Harm to the user's devices or work	Lake uses so much memory, processing time or power on a device that the user's other applications crash, unsaved work is lost, or the device itself crashes or freezes.

3 System Boundaries and Components

3.1 Coordinator

This component exists on all nodes, but the following responsibilities are only run on the leader node:

- Providing an OpenAI-API compatible API to the user for use with tools like OpenCode.
- Tracking cluster health, and detecting node failures.
- Notifying nodes of partitioning changes.
- Rebuilding the cluster and saving in-progress requests in the case of node failure, removal, or addition.
- Measuring node connection performance.

On a follower node, it is primarily responsible for managing the inference engine on its device, following the instructions of the leader node.

3.2 Model Partitioner

This component determines how to partition the model among the participating devices. It decides at which points to cut the graph, how large each partition is, and which device gets each partition. This component is integral to ensuring that each device is assigned an appropriate partition given the device's computational and networking performance.

3.3 Inference Engine

This component is the high-performance inference engine running on each device. It will handle execution of the given model partition for each device. It will call into ggml using the Rust-to-C FFI (TODO: glossary) to control the execution of a device's model partition.

3.4 Tensor I/O

This component handles the compressing, sending, receiving, and decompressing of tensors between devices. It will leverage gRPC for high-performance communication, with the potential to use a custom solution should gRPC fail to meet our requirements in practice.

3.5 Boundaries

The system boundaries for Lake primarily consist of:

- The Coordinator's OpenAI-API compatible API.
- Model files (GGUF) loaded from disk.
- Networking components between devices (Thunderbolt, Ethernet, Wi-Fi, routers).
- The OS and GPU resources on each individual device.

4 Critical Assumptions

We assume:

- The model file (GGUF) provided to Lake is already a functional LLM before partitioning.
- User requests are only made through the API on the leader device.
- Only one model will be run by the pool at a time.
- The local area network connecting the devices is private, and trusted by the user.

5 Failure Mode and Effect Analysis

The following Failure Mode and Effect Analysis (FMEA) table summarizes potential failure modes for the Lake system based on each component. It also details potential causes and effects for each of the failures, along with a recommended action to mitigate the potential failure. Components are listed based on their appearance above in Section 3. Beyond this, there are also hazards listed for general communication failures that may take place in between nodes or components, as well as privacy failures.

Component	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Reference
Coordinator	Incorrect or outdated device health statistics are reported back to the user.	Coordinator sends requests to device that is unable to handle requests and inference fails.	Health checks are too infrequent from the coordinator or user interface displays older data than available.	Scheduled health checks to each device and removal of data from user interface as soon as new data reaches leader node.	SR0-1	H0-1
	Coordinator component in leader node fails to notify follower node of partitioning change.	Follower node sends data to other failed follower node and inference fails.	Lack of proper and timely notification system and health checks from leader node.	Scheduled health checks to each device prior to each message to ensure inference is not lost on a failed node.	SR0-2	H0-2
	Single device node is allocated 100% of the model.	Fails the core purpose of Lake and overworks single device by avoiding parallelization.	Improper boundaries on model partition algorithm.	Boundaries to ensure each device is allocated >0% of the model, and rebalancing checks made throughout a session to verify boundary is not violated.	SR1-1	H1-1
	The model is insufficiently partitioned and cannot be split up amongst the device pool. (# Partitions < # Devices)	Model runs on only part of available hardware and does not achieve best possible performance.	Poor constraints on model partition algorithm.	Ensure that layers are split up into enough pieces to span all devices (pigeonhole principle) and build checks into initial algorithm to guarantee each device handles part of inference.	SR1-2	H1-2

Model Partition						
	Device is allocated a partition larger than it's available memory.	Device crashes and loss of network.	Improper consideration of device hardware requirements and lack of tolerances.	Ensure tolerances built into partition algorithm to keep models at (100-t)% of device availability (t is specified threshold that will be determined in testing).	SR1-3	H1-3
Inference Engine	Follower device fails or crashes during local inference.	Inference is lost and response cannot be returned to user.	A myriad of local failures could cause the device to not complete the inference. This could be hardware constraints or other device processes interfering.	Leader device is able to prevent failures from being shown to user and send retry request(s) to produce proper output.	SR2-1	H2-1
	Node receives incoherent tensors and cannot produce output to user.	User does not receive a response to their prompt and cannot use the system.	Communication failure (lost or corrupted data) between two nodes.	Place checks throughout the inference process for corrupted or lost tensors and restart inference from leader, temporarily removing the node where corruption/loss occurred from the network. Additionally, checksums can be implemented to catch network corruption, with the system restarting if the checksums do not match.	SR3-1	H3-1

Tensor I/O	Leader node fails to extract compressed tensors.	User does not receive a response to their prompt and cannot use the system.	*i* - Follower device fails to produce proper output or *ii* - Leader cannot interpret output and run final inference to produce output.	Using the checks identified in H3-1, identify the node causing the failure and temporarily remove it from the network before restarting. If no problem is indicated from any follower node, restart inference and notify user of the issue.	SR3-2	H3-2
	Communication between two nodes times out between devices or fails entirely.	Failure of system, loss of inference if midway through session.	Failure of communication protocol between nodes at hardware level (Thunderbolt, Wi-Fi, Ethernet).	Monitor average communication times between nodes and alert user to check connections if it exceeds a threshold. This can be measured through packet loss and throughput metrics. Restart inference if communication is lost and remove failed node.	SR4-1	H4-1
	Device health status communication fails.	Leader node unaware if device fails, loss of system is possible if unknown device's performance is degraded.	Failure on behalf of leader node to send outbound requests, or failure of follower nodes to receive and respond to requests.	Monitor for devices not returning health checks, after multiple missed, remove device from inference network until health response is received by leader. In the case of leader node failure, the system's follower nodes will interpret a lack of incoming health requests as leader failure and shut down since no further coordination can take place.	SR4-2	H4-2

General Communication	Communication becomes noticeably slower for user compared to expected performance.	Users leaving the system and opting for another provider.	Inefficiency in between nodes or complete network slowness. Model is split up across too many devices causing too much communication overhead for each prompt.	Simplification of network. If inference is taking too long to complete based on historical system data, remove least capable devices on the network to reduce communication overhead.	SR4-3	H4-3
Privacy	Untrusted node on the device network.	Allows unintended access to the system and potential loss of data to bad actors. Reduces user trust in the system and may lead to loss of users.	Lack of initial authorization of each node or operation of the system on an untrusted network where unknown devices can join.	Ensure that system is only operated on limited networks where all potential devices are known. Alternatively, the leader node must gain approval from the user before adding an unknown device to the network.	SR5-1	H5-1
	Data leakage from a node leading to untrusted access or data leakage.	Untrusted users who have access to any of the participating nodes can see and steal data. Reduces user trust in the system and could lead to degraded or failed performance.	Lack of initial authentication, or storage of unnecessary temporary data on intermediary devices.	Ensure that any devices in the network are explicitly approved by the user and that any unnecessary data is not stored on intermediary devices.	SR5-2	H5-2

6 Safety and Security Requirements

- **SR0-1** – The system shall identify and discard stale device health data from determining device availability by configuring a time-to-live for valid data.
- **SR0-2** – The system shall only route inference requests to nodes that currently have a confirmed model layer partition.
- **SR1-1** – The system shall assign at least two nodes for model layer partitioning during distributed inference.
- **SR1-2** – The system shall ensure all model layers are assigned to a node, and that each device configured to run model layers receives a non-zero model layer partition assignment.
- **SR1-3** – The system shall reject models and layer allocations that are estimated to exceed the memory capacities of the Lake network or individual nodes, including a configured safety margin. Related SRS requirements: FR.S.2.1, FR.S.2.2.
- **SR2-1** – The system shall attempt an automatic bounded recovery and retry on inference requests if a follower node fails during inference. If the follower node is unable to be recovered within the configured recovery limits, the system must report to the API that the request has failed. Related SRS requirements: FR.S.1.10, FR.S.1.11, FR.S.1.13.
- **SR3-1** – The system shall identify corrupted and incomplete tensors throughout the distributed inference with integrity checks and communication timeouts to discard affected data and retry requests.
- **SR3-2** – The system shall log failed tensor decoding metadata and report a failure to the coordinator so the inference request can recover, subject to SR5-2.
- **SR4-1** – The system shall configure a timeout for communication between nodes, and handle exceeded timeouts according to the appropriate recovery or failure policy.
- **SR4-2** – The system shall mark a node as unavailable for inference or model layer partitioning when the node health criteria are not met, and require a successful health check before marking the same node as available. During a leader node failure, follower nodes shall detect and initiate a shutdown. Related SRS requirements: FR.S.1.24 and FR.S.1.25.
- **SR4-3** – The system shall utilize configured values or historical data to determine a baseline inter-node communication latency and notify the coordinator when that baseline is exceeded by a specified tolerance.
- **SR5-1** – The system shall require user approval on the leader node to authorize a follower node, and user confirmation on the follower node to confirm the authorization before it can participate in the Lake network. Related SRS requirements: FR.S.1.17, FR.S.1.18, FR.S.1.20, FR.S.1.21, NFR.S.12.
- **SR5-2** – The system shall not record prompts, responses, tensor data, and KV cache content in diagnostic logs. Related SRS requirements: NFR.S.10.

7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

8 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

8.1 AI Tools Used

- Claude Sonnet 5.5

8.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

Claude Sonnet 5.5 was used for assistance in creating the FMEA table formatting (no content). It was used for creating the landscape table with custom column widths that could span multiple pages.

8.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

9 Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?

4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?